

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I C Sangita(192373059)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

csangita

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.

b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

Solution:

Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message:

Ciphertext: GSRH RH Z HVXIVG

The structure of the text suggests a Monoalphabetic Substitution Cipher, where each plaintext letter is replaced with a fixed ciphertext letter according to a substitution alphabet.

- a) A Monoalphabetic Substitution Cipher replaces each letter of the plaintext with another unique letter throughout the entire message. The substitution remains fixed for all occurrences of a letter.

Substitution Table:

Plain Text	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher Text	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Decryption Process:

Ciphertext: GSRH RH Z HVXIVG

Using the substitution table:

Cipher table	G	S	R	H	R	H	Z	H	V	X	I	V	G
Plain letter	T	H	I	S	I	S	A	S	E	C	R	E	T

Recovered Plaintext:

THIS IS A SECRET

b) Pattern Analysis:

The intercepted message contains four separate words:

GSRH RH Z HVXIVG

The second word consists of two letters and appears only once. After decryption, it becomes "IS", which is one of the most common two-letter English words.

The third word contains only one letter:

Z

After substitution, it becomes "A", which is a common one-letter English word.

The first word: GSRH decrypts to: THIS

Which is a meaningful English word and fits naturally at the beginning of a sentence.

The last word: HVXIVG decrypts to: SCERET

Forming the complete sentence: THIS IS A SECRET

Cipher Characteristics

The same ciphertext letter always maps to the same plaintext letter throughout the message.

Examples:

- $G \rightarrow T$
- $H \rightarrow S$
- $R \rightarrow I$
- $V \rightarrow E$

This fixed one-to-one mapping is the main characteristic of a Monoalphabetic Substitution Cipher.

Since the substitution rule does not change during encryption, repeated letters in the ciphertext always represent the same plaintext letters. This property allows attackers to use frequency analysis, common word patterns, and repeated letter structures to recover the original message.

Finally, the decrypted sentence is meaningful, grammatically correct, and follows common English language patterns. The successful reconstruction of the message confirms that the ciphertext was generated using a monoalphabetic substitution technique based on a fixed substitution alphabet.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement.

A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

a. Determine the most probable original message by reversing the encryption process.

b. Critically evaluate why such a method is vulnerable in modern communication systems.

c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Solution:

Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company uses a Rail Fence Cipher (3 Rails), which is a classical Transposition cipher.

The received encrypted message is:

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN

a) The Rail Fence Cipher is a transposition cipher that encrypts a message by writing the plaintext in a zig-zag pattern across a fixed number of rails and then reading the characters row by row. Unlike substitution ciphers, the letters themselves are not changed; only their positions are rearranged.

In this scenario, the message has been encrypted using 3 rails.

Decryption Process

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

Since the encryption uses 3 rails, the ciphertext is arranged back into the rail pattern and then read in a zig-zag manner.

Rail Arrangement

Rail Arrangement

Step 1: Count the letters

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN

There are 25 letters.

Step 2: Draw the zig-zag pattern for 3 rails

Rail 1: * * * * *

Rail 2: * * * * * * * *

Rail 3: * * * *

The positions are:

- Rail 1: 1, 5, 9, 13, 17, 21, 25 → 7 letters
- Rail 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24 → 12 letters
- Rail 3: 3, 7, 11, 15, 19, 23 → 6 letters

Total = 7 + 12 + 6 = 25 letters

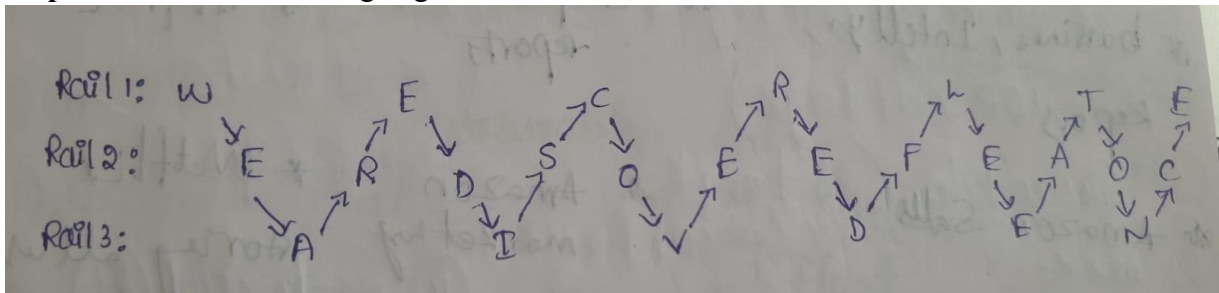
Step 3: Divide the ciphertext

Rail 1 (7 letters): WECRLTE

Rail 2 (12 letters): ERDSOEFEAOC

Rail 3 (6 letters): AIVDEN

Step 4: Place them in the zig-zag



Reading the characters in the zig-zag order reconstructs the original message:

WEAREDISCOVEREDFLEEATONCE

Adding spaces to improve readability:

WE ARE DISCOVERED FLEE AT ONCE

- b) The Rail Fence Cipher provides only basic security because it rearranges the positions of characters without changing the letters themselves. This makes it vulnerable to several types of cryptographic attacks.

Reasons for Vulnerability

- The letters of the plaintext remain unchanged, making frequency analysis easier.
- The number of possible rail values is very small, allowing attackers to perform brute-force attacks quickly.
- Modern computers can try all possible rail combinations within a few seconds.
- No secret cryptographic key is used apart from the number of rails, resulting in a very small key space.
- The cipher does not provide authentication or integrity, so attackers can modify the message without detection.

Due to these weaknesses, the Rail Fence Cipher is considered unsuitable for protecting confidential corporate communication.

- c) A practical improvement is to replace the Rail Fence Cipher with the Advanced Encryption Standard (AES-256).

Enhanced Communication Process

Plaintext



AES-256 Encryption



Encrypted Ciphertext



Secure Transmission



AES-256 Decryption



Original Plaintext

AES-256 is a modern symmetric encryption algorithm that uses a 256-bit secret key. It provides strong confidentiality, a large key space, and resistance against brute-force attacks. Even if an attacker intercepts the encrypted message, it cannot be decrypted without the correct encryption key.

Finally, the decryption process successfully reconstructs the original message from the ciphertext, demonstrating how the Rail Fence Cipher works. However, the analysis shows that the cipher offers only minimal security because it relies solely on rearranging characters. Replacing it with a modern encryption algorithm such as AES-256 significantly improves confidentiality and protects sensitive corporate communication from modern cryptographic attacks.